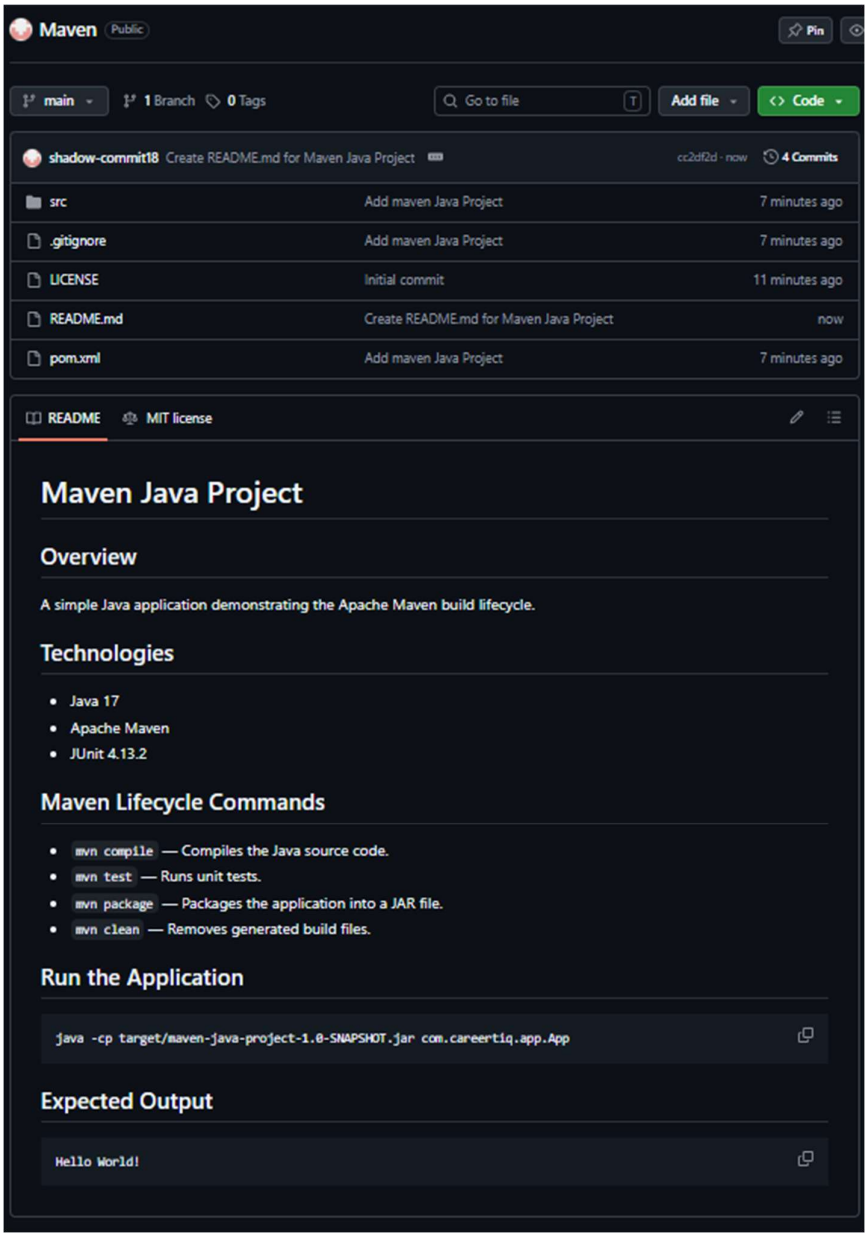
Name – Sairaj Dattaray Bagal

Subject - Devops

Department - AIML SY

Assignment – Maven



Apache Maven Lifecycle Exercise

1. Aim

To configure and use Apache Maven for a Java project, understand the Project Object Model (POM), execute Maven lifecycle phases, and run the compiled Java application.

2. Software Requirements

- Operating System: Windows 11
- Java Development Kit: JDK 17
- Apache Maven: Installed and configured
- Command Prompt (CMD)

3. Procedure

1. Verified Java and Maven using java -version and mvn -version.
2. Generated a Java project using the Maven Quickstart archetype.
3. Inspected the project structure and configured pom.xml.
4. Compiled the source code using mvn compile.
5. Executed unit tests using mvn test.
6. Packaged the project using mvn package.
7. Removed generated build files using mvn clean.
8. Executed the packaged Java application.

4. Maven Commands

- mvn compile — Compiles the main Java source code.
- mvn test — Executes the project's unit tests.
- mvn package — Runs the preceding required phases and packages the application into a JAR.
- mvn clean — Removes generated build output from the target directory.

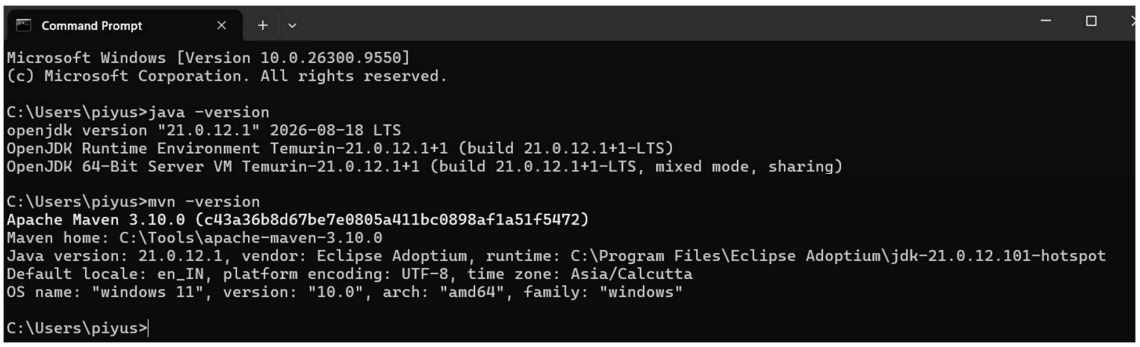
5. Result

The Maven project was configured and packaged successfully. The generated JAR was verified. The application execution result is documented using the attached terminal screenshot.

6. Conclusion

The exercise demonstrated how Apache Maven manages Java project configuration, compilation, testing, packaging, and cleaning through its standard build lifecycle.

MAVEN and JAVA versions verification



File Structure



POM Configuration

pom.xml

C: > Users > piyus > maven-java-project > pom.xml

```xml

<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.careertiq.app</groupId>
    <artifactId>maven-java-project</artifactId>
    <version>1.0-SNAPSHOT</version>
    <packaging>jar</packaging>

    <name>maven-java-project</name>
    <url>https://maven.apache.org</url>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    </properties>

    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>4.13.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

</project>

```

Compile and Test with verification

```
PS C:\Users\piyus\maven-java-project> java -cp target\maven-java-project-1.0-SNAPSHOT.jar com.careertiq.app.App
Hello World!
PS C:\Users\piyus\maven-java-project>
```

```
PS C:\Users\piyus\maven-java-project> mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------< com.careertiq.app:maven-java-project >----------------
[INFO] Building maven-java-project 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24072 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ maven-java-project ---
[INFO] skip non existing resourceDirectory C:\Users\piyus\maven-java-project\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ maven-java-project ---
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Nothing to compile - all classes are up to date.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.985 s
[INFO] Finished at: 2026-10-13T18:08:51+05:30
[INFO] ------------------------------------------------------------------------
```

```
PS C:\Users\piyus\maven-java-project> dir target\classes\com\careertiq\app


    Directory: C:\Users\piyus\maven-java-project\target\classes\com\careertiq\app


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         13-10-2026     18:00            549 App.class


PS C:\Users\piyus\maven-java-project>
```

```
PS C:\Users\piyus\maven-java-project> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------< com.careertiq.app:maven-java-project >----------------
[INFO] Building maven-java-project 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24072 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ maven-java-project ---
[INFO] skip non existing resourceDirectory C:\Users\piyus\maven-java-project\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ maven-java-project ---
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ maven-java-project ---
[INFO] skip non existing resourceDirectory C:\Users\piyus\maven-java-project\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ maven-java-project ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.6.0:test (default-test) @ maven-java-project ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.careertiq.app.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.027 s -- in com.careertiq.app.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.024 s
[INFO] Finished at: 2026-10-13T18:09:36+05:30
[INFO] ------------------------------------------------------------------------
```

Clean Lifecycle

```
PS C:\Users\piyus\maven-java-project> mvn clean
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------< com.careertiq.app:maven-java-project >----------------
[INFO] Building maven-java-project 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24072 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ maven-java-project ---
[INFO] Deleting C:\Users\piyus\maven-java-project\target
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.499 s
[INFO] Finished at: 2026-10-13T18:10:00+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\piyus\maven-java-project>
```

Package and JAR

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> cd C:\Users\piyus\maven-java-project
PS C:\Users\piyus\maven-java-project> mvn package
[INFO] Scanning for projects...
```

```
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.1:jar (default-jar) @ maven-java-project ---
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.491 s
[INFO] Finished at: 2026-10-13T18:06:59+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\piyus\maven-java-project>
```

Application Output

